

THE BRITISH
JOURNAL OF PHOTOGRAPHY.

No. 613. Vol. XIX.—FEBRUARY 2, 1872.

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A CHARACTER OF CERTAIN PRESERVATIVE FILMS.

HERR Schultz-Sellack has recently called attention to a remarkable agreement in certain properties between chemically-clean glass surfaces and photographic films coated with tannin or albumen. The particular property in which they appear to agree is this—that an albumenised or tanned surface, when breathed upon, takes the moisture evenly and loses it gradually by evaporation, just like a clean glass plate.

This wonderful discovery (!) is announced in *Poggendorf's Annalen*, and a notice of it appears in the last *Journal of the Chemical Society*. If Herr Schultz-Sellack had endeavoured to make himself acquainted with the literature of photography he would scarcely have taken the trouble to publish the statement of a fact well known to all who have much experience in dry-plate work. A film protected by albumen, tannin, or any other substance capable of absorbing moisture, resembles a perfectly-clean glass plate in the mode of condensation of moisture and the manner in which an aqueous film disappears from the surface: but this property of a somewhat hygroscopic surface is one which might be easily anticipated and has long been observed.

“Whether this hygroscopic property is beneficial or otherwise we are scarcely in a position to decide, as wide differences of opinion exist amongst practical men on this subject—on the one hand, Mr. M. Carey Lea considering nearly complete desiccation of a dry plate conducive to greater sensitiveness, and on the other, a large number of operators declaring that a preservative capable of keeping the plate in a semi-moist condition is most advantageous. Our own experience is in favour of Mr. Lea's position; but it is by no means improbable that some of the so-called preservatives act best when moist, and others when the film is fully dried. We shall content ourselves with citing a single case in point.”

When a film of iodide of silver is washed free from extraneous matter, and then covered with a solution of ferrocyanide of potassium, a very sensitive layer is obtained while the film is moist, but if dried fully the action of light upon the surface is very slow. Suppose, however, that we add to the ferrocyanide solution, previous to its application to the iodide film, a quantity of honey, a little glycerine, or a very minute amount of nitrate of magnesia, a comparatively sensitive film is obtained, which, though apparently dry, is still not completely so. Here, then, is a remarkable case, parallel with the analogous action of nitrate of silver moist and dry upon sensitive layers.

A SHELLAC VARNISH.

IN our last volume there appeared two very interesting notes—one by Mr. G. Watmough Webster, F.C.S., and the other by Mr. A. R. Brown—on the preparation of a shellac varnish of a peculiar kind, and easily miscible with water. Though we have been long familiar with the mode of preparing such a varnish—thanks to the kindness of a friend—we have lately gone over the whole matter, and have been so interested that we now venture to recall the matter to the consideration of our readers—this week mentioning the most convenient mode of preparing the so-called varnish, and reserving to another time an account of several applications of it which appear to possess some interest.

The most rapid mode of obtaining the shellac varnish conveniently is certainly that of Mr. Webster. This gentleman dissolves the resin in spirit. We may remark that strong spirit gives the most satisfactory results. The solution when now treated with a very small amount of the ordinary liquid ammonia becomes immediately miscible with water. Care must be taken, however, only to add sufficient ammonia to accomplish the desired end, and this can be easily accomplished by testing a little of the mixture by addition of water after every fresh dose of alkali. In this way any unnecessary deepening of the colour is avoided, and a useful liquid obtained.

We now turn to the second plan—that mentioned by Mr. Brown. It has been long well known that *fused* shellac dissolves much more easily in alcohol than the resin which has not been melted. Mr. Brown adopts the same plan in obtaining solution in ammonia. The resin is melted under a layer of water, and, when perfectly fused, strong liquor of ammonia is slowly poured in, the whole being carefully mixed. Gradually the resin is taken up by the liquid, and a good, brown mixture obtained. If too much ammonia be used here the colour is greatly deepened, owing to the action of the alkali on a peculiar impurity in the resin.

Both the above processes work well; but, when small quantities have to be prepared, Mr. Webster's plan yields the best product, as a solution of shellac in spirit can be obtained having a comparatively slight colour, whereas the direct use of ammonia deepens the tint. Whichever plan be used a good result can be arrived at, and a liquid obtained capable of numerous and most useful applications, whether as a species of varnish, a cement, or material for, in a sense, waterproofing paper.

CHROMATE of potash is used in dyeing in conjunction with dextrine, &c., for discharging the colour from various fabrics; it seems, however, that a difficulty has been noted in using such a mixture, for it is found that certain of the goods become discoloured by the chromate, and that this undesired colour is not easily removed. Will our readers be "surprised to hear" that the cause of this circumstance appears to have been only just found out, if we may judge from a quotation in the *Journal of the Chemical Society*? It has been discovered that it is exposure to light during treatment which prevents the removal of the offending material. This circumstance indicates how little attention is in general paid to the phenomena attending the chemical action of light; and, notwithstanding the fact that the remarkable property possessed by the chromates of rendering insoluble gelatine, &c., has been utilised to an extraordinary extent in the various pigment printing processes, operative chemists seem to have only now become alive to the fact, and able to apply the obvious remedy in using yellow glass for the rooms in which the treatment of goods with the chromated dextrine is carried on.

BLISTERING OF ALBUMEN SURFACES.

HAVING read with much interest the various remarks which have appeared in your columns of late on the blistering of albumenised surfaces, I will, with your approbation, state my conviction of the cause.

First, let us examine the blisters and get at a clear notion of their physical aspect.

All who have spoken on the subject appear to agree that the more highly a paper is albumenised the more do we suffer from the annoyance. Well, such a strong film gives us the opportunity to detach some of those large blisters which have been formed by the coalescence of a number of small blisters.

Proceeding after the manner of a microscopist in his dissection of animal structure, we depress the sheet or photograph to the bottom of a dish with water, and there introduce a sharp-pointed knife, whereupon air escapes. Here we simply note this fact, and continue the dissection. Having detached a vesicular film, and caught it on a small plate of glass with the innermost side upwards, it will be found in almost every experiment to have the surface fibre of the paper attached, thus showing that it does not part from the surface of the paper as a film pure and simple. We therefore dismiss from our minds the idea that the outer surface only had been rendered insoluble, while the inner portion had been dissolved; were this the case the blister would be *filled with the albumen solution*.

In accordance with the law discovered by Graham—that a colloid solution, such as albumen, cannot pass through a colloidal septum—the blisters cannot be the *result of expansion*. Further: where a colloid septum is wetted on both sides, liquids of the nature of water can diffuse through, and therefore the blisters would be filled with water or solution of crystalloid salt, such as hyposulphite of soda.

Now we must search deeper and wider for the cause. If I am not mistaken, the blisters show themselves in greater force, in this country, from the middle of May to the end of July. During this period the natural waters are of the most favourable temperature for the absorption of air or gases on which the aquatic plants feed; gases coming in contact with or passing through the cells of the living plant are decomposed, and the residual gas discharged into the water. As the water becomes lowered in temperature it is unable to hold these absorbed gases, the minimum temperature being from 32° to 40° Fah.

It is from this cause, in my opinion, that ice floats on the surface. As water becomes lower in temperature minute bubbles of air are seen to grow, as it were, on submerged objects; eventually the buoyancy overcomes the force of adhesion and gravity, whereupon the air-beads rise and become entangled amongst the water crystals—ice—and thus render the ice lighter, bulk for bulk, than the water from which it has been formed.

Some of my readers will, most likely, say—“What has this to do with the blistering of albumen surfaces?” Well, let us see how these ideas, derived from anterior observations, assist us in the matter.

There are many substances beside the growing tissue and cells having the property of condensing gases in their pores, and often entering into combination with such avidity that sufficient heat is liberated to render the substance incandescent. Such is the case when hydrogen gas is impinged on platinum in a spongy form. Fresh-burned charcoal is another substance having the property of condensing many times its own volume of gas, and by mutual attraction to the occluded air the external air sticks as a film on the substance with considerable force—wets the surface, if I may use the simile. Most substances possessing extremely minute pores or interstices have this property, in a more or less degree, for different kinds of gases; therefore we may fairly class the fine texture of photographic paper amongst such bodies.

Grant that there is an analogy between the above-mentioned cases, then the cause and effect become intelligible. First we have a hard-sized paper with gases or air condensed in the interstices, and with the external air adhering to it; in this condition it is coated with an amorphous film of albumen. All appears to go on favourably until the fixing process; here the

temperature is reduced by the addition of water contained in the prints. Now the absorbed air which the water contains finding a nucleus on its permeating the body of the paper grows immediately (as I pointed out when directing your attention to the phenomena proceeding from the freezing of the water) in the surface-fibre, thus it is likely that the blisters are formed in all cases where air is enclosed. Hyposulphite of soda when entering into solution reduces the temperature considerably—the air being then held with so little force readily unites with a nucleus.

To counteract this natural effect it is obvious that means should be taken to prevent the lowering of the temperature on the addition of water to the hypo, solution, or on the transference of prints from the hypo, into water containing absorbed air.

If this be unmanageable, the easiest thing to do is to dip the prints into methylated alcohol, after leaving the toning bath. This treatment removes the sizing from the paper, and the prints appear translucent while wet. They should now be *well washed*; any disengaged air will then be able to escape through the porous body of the paper. It has been recommended to employ a large charcoal filter to absorb the air or gases, but, like all filters, it will require cleaning out, and the charcoal needs re-burning.

A word on filters in conclusion. The public purchase filters, and imagine that they are to have pure water, or clarified solution, for evermore without trouble; let them be undeceived, because when charcoal becomes saturated it must be re-burned to restore its purifying properties.

F. W. HART, F.C.S.

COLLODIO-BROMIDE.

IN last week's issue your idea of combining silver with a new base in the collodio-bromide is so good that I think it just as well to mention that the very notion of acetate of silver has been carried out by me a dozen years ago, although in another form. I made a collodion for wet-plate work in which an iodide was used with about an equal part of acetate of soda; the collodion was accelerated to such a degree that my exposure was reduced to about a fourth of what I would have given without the acetate. The plate developed beautifully and free from fog. Here, again, comes in the law of compensation; my thoroughbred collodion was, at the end of a dozen hours, as slow as a team of oxen, as vitality seemed to expend itself in one *coup d'état*, and the second day it was capital—for cleaning plates.

A bromo-iodised collodion could not be stimulated with any such exertion. Since practising the collodio-bromide process I have often promised myself an experiment in this direction; but, somehow or other, partly from my failure when bromide was in the collodion, the trial has ended in the intention only.

Mr. Henry Cooper's success with lactate of ammonia has inspired me with a strong hope that collodio-bromide is to be the perfection of "everything to be desired" in dry-plate work.

I am glad to learn that Col. Wortley has found out the conditions of making a bromised collodion which arrives at maturity without passing through that infantile existence which I have always regarded as unfavourable to the best results. When I made the statement in your Journal, it was from the united experience of all my collodio-bromide acquaintances, which, if pinned together, would, in respect of mere lapse of time, extend over the greater part of a century.

Although bromised collodion will give good results when freshly made, the development of the image is almost a matter of physical force—more

ammonia, more silver, and such like—in order that the excellent prints expected from your exertions may be superb indeed, in consequence of the labour bestowed on the negative.

I do not condemn Col. Wortley's mode of working, for I have lately seen some of his negatives possessing the very highest qualities attainable by any process; but, in support of my own recommendation that the collodion should be left to ripen as a means of lightening our labours in the dark room, I may state that for over two years I never used silver as an intensifier, alkaline development giving me always the right density, and occasionally too much. With an "infant" collodion silver is the only "soothing syrup" that affords nourishment to the sickly image.

J. W. GOUGH.

THE FADING OF MERCURY-TONED TRANSPARENCIES.

IN the multitude of counsellors there is said to be wisdom. This may be true enough in the abstract, but in its special applications it may sometimes lead to bewilderment, more particularly if the counsel of one of the “multitude” be diametrically opposed to that of another counsellor.

In looking over the pages of your useful Year-Book and ALMANAC I find that Mr. Brooks has something to say in behalf of a process that has of late received some rude words and rough treatment at the hands of the Editors and others, the process to which I refer being the toning of transparencies by means of mercury. As Mr. Brooks has issued a great number of photo-crayons toned with this agent, it would appear that he felt somewhat uneasy at the prospects of their fading; but he says—“I now feel quite decided that, if the picture be protected by a good varnish, fading will not ensue when a picture is thoroughly toned with mercury.”

Let not Mr. Brooks nor any one else lay this flattering unction to his soul. If varnish be the ægis which is trusted to ward off the destructive effects of fading from the picture, I warn those interested from putting any trust in it; for I have in my possession several of such pictures varnished in the most faultless manner, with varnish of unexceptionable quality, which have faded away into almost invisibility. Now, as these transparencies have been prepared by myself as well as by photographers of recognised high position, including the introducer of the photo-crayon, I believe that I am warranted in assuming that every requisite care has been taken both in the toning and the washing.

I am aware that it has been said that heat is an all-important element in the destruction of a mercury-toned glass picture, and that from this cause the lantern pictures are peculiarly liable to fade, whereas those not subjected to heat will stand. I regret to be compelled to shatter the prop that is raised from this basis to support the stability of the photo-crayon; for in a

series of two dozen mercury-toned lantern pictures, purchased not much over two years ago, several which have never been in the lantern or exposed to heat in any way have faded into worthlessness.

I think, therefore, from what I have said, the truth of which the Editors will attest—seeing that I send them herewith several specimens—there is only one lesson to be drawn, and it is this:—Above all things, as you value your reputation, avoid the use of mercury for toning either lantern transparencies or photo-crayon pictures *à la* Sarony.

DON QUIXOTE.

NOTES ON PASSING EVENTS.

BY A PERIPATETIC PHOTOGRAPHER.

THE *Attractive Chemical Experiments for Youthful Readers*, by the Editor of the ALMANAC, and which, I suppose, was “crowded out” of that widely-circulated annual, forms a special kind of reading that will prove attractive to old as well as youthful readers; for I believe I am quite safe in affirming that chemical “tricks” are equally appreciated by old and young. I recollect attending on one occasion an entertainment of a very attractive nature, which may be described as follows:—The lecturer, having placed on the table several glasses apparently quite clean and empty, proceeded to fill them from an ordinary decanter full of water; first, however, asking the audience to name the liquors they preferred. Suffice it to say that by request the glasses were respectively filled with what purported to be port, sherry, whisky, rum, brandy, and so forth. When milk was asked for, the white fluid was immediately produced; so with soda water, which fizzed and sparkled, and with ginger beer, the froth of which stood stiff, without subsiding. One gentleman, anathematising the temperance fluids, asked for a glass of something hot and fiery; and, in response, a glass was filled and from it issued flames of fire. As the clear glass decanter was filled with water in the presence of the audience this trick was rendered more surprising than if an opaque bottle had been employed, as is the custom with the majority of “inexhaustible bottle” conjurers. As this is not written for youthful readers, but for those who are everyday readers of the Journal, I will pay a compliment to their chemical perspicuity and assume that they know how these changes took place. However, if I hear that I have assumed too much, I will next month give the explanation of these small chemical marvels.

Who does not sympathise with a “Black-Country Foto.,” who appears to think that there is no use in his sending portraits to the London exhibition because he can’t afford to have his negatives worked upon to the degree that

seems requisite to permit of successful competition? If the gossip that was indulged in by the visitors to the exhibition is to be credited, the preparing of the portrait negatives of one amateur must have been attended with no small cost; for it was asserted that he not only had skilled professional aid, but that the negatives were handed over to a retoucher to be dealt with regardless of expense. Where Dives goes in for honours Lazarus has, under similar circumstances, but small chances of success. When specimens from untouched negatives are exhibited (not necessarily to the exclusion of prints after touching), then will some grumbling cease.

The South London Photographic Society has at last quitted its old quarters in the City of London College and has removed to a more central situation. Between the convenience and elegance of Arundel Hall and the City of London College there is no comparison; and the removal ought to tell most beneficially upon the prospects of the society.

The communications already made to photographic societies this year indicate activity. At the recent meeting of the London Photographic Society Mr. Sawyer read a paper on *Photography in the Printing Press*, in which he took occasion to discuss the novelty of Mr. Edwards's process, so far as its patentability was concerned. At the monthly meeting of the South London Society there were two papers—one by Mr. Dunmore, on *The Cause and Prevention of Blisters on Albumenised Paper*, the other being by Mr. Croughton, *On Photographic versus Literal Truth*. In the former the cause of blistering was alleged to be the imperfectly rendering soluble of the under surface of the albumen; in the latter (the discussion on which is deferred till next meeting) the essayist spoke of the exaggerations and distortions caused by using a short-focus lens, and he condemned the pandering to human weakness displayed by a large firm who owed its success to the amount of flattery bestowed upon its customers by the inordinate use of retouching. At the meeting of the Edinburgh Photographic Society a *resumé* of the progress of the past year was given by Dr. Nicol, who is of opinion that the conversion of a distorting lens into a non-distorting one is an improvement of doubtful value—an opinion in which the generality of photographers will scarcely coincide. The Manchester Photographic Society had a paper by Mr. Coote *On the Good Keeping*

Qualities of Collodio-Albumen Plates between Exposure and Development.

It is an unfortunate thing that many dry plates, after exposure, do not retain the latent image; hence the necessity for developing as quickly as possible after the camera has effected its portion of the work. As a type of processes of this kind the tannin may be mentioned, whether this substance be used as a preservative in the silver bath or the emulsion methods. Mr. Coote developed successfully a collodio-albumen plate which had been kept for more than two years after being exposed.

TRANSALPINE PHOTOGRAPHY.

I HAD been out of sorts all the spring; my business engagements had kept me longer than usual in town, and prevented my getting a little relaxation. It was not until the end of August I was able to put into execution a long-contemplated visit to some relatives in Italy, and try what change of air and a holiday would do for me.

So away we started—a party of four, my wife and two ladies under my charge—and armed with Cook's tickets we made our way *via* Newhaven to Paris, which we reached late at night. Next day, as our train for Italy did not leave till the evening, we wandered about Paris, visiting the sad scenes of Communistic destruction. Plenty of photographs in every shop illustrative of the wretched devastation which was visible on all sides. In a place like Paris there was of course all qualities, shapes, and sizes; and, in addition to *souvenirs* of the ruined public buildings, there were to be bought a great variety purporting to be faithful representations of the events of the siege, such as barricades, executions, streets with dead bodies lying in them, and so forth. These, it is needless to say, were fictitious, carefully-arranged tableaux, but terribly natural, and greedily bought by tourists, the greater portion of whom never for one moment questioned the reality of the representation.

At night we started from the Lyons station for our long-through journey over Mont Cenis, for the tunnel was not yet complete. By morning we were travelling through the lovely scenery from Chambery to St. Michel, the commencement of the Fell railway over the mountain. Here there is a splendid field for the photographic artist, and, so far as I could discover, hitherto unworked. As the railway wound its tortuous way amidst the charming mountain scenery the landscape changed at every turn—bridges, cottages, foliage of every variety, rivers, brooks, and torrents coming into

view every instant. Here a photographer might spend weeks without exhausting the rich field for his camera.

At three o'clock St. Michel was reached, and after some inexplicable delay—which, however, gave us the opportunity of making a comfortable dinner—we changed from the ordinary line of railway to the special narrow-gauge carriages on the Fell system for crossing the mountain. A grand journey it is—zigzagging over the steep ascent, through scenery of charming and varied character! This Fell system has been so often described I shall forbear to say more about it, except that it contrasted very favourably both in time and comfort with the old diligence journey I had undergone some years before over the same route.

From our delay in starting we got to the other side of the mountain just in time to miss the last train for Turin, which we had the satisfaction to see leaving the station as we entered it. This involved our stopping at Susa for the night; and, of all wretched, miserable places, commend me to Susa. Nobody stops there who can get away. It was eleven at night, and we had been travelling ever since 8.40 the previous evening. Telling the porter to take us to the best hotel, away we trudged, only too glad to take advantage of any resting-place for the night we could get—a regular old-fashioned Italian hotel, the entry through the kitchen, and the lower part of which, built round a courtyard, was devoted to the horses, cows, and pigs—very picturesque, no doubt, but, as may be easily imagined, anything but savoury in the way of scent.

The next morning we explored the town, which, though small, is the seat of a bishopric. It contains plenty of quaint bits of buildings, which would make good food for the camera, but not a photograph did I see for sale. As soon as breakfast was over we set off for Milan, being anxious to reach that city, as we expected some of our family to meet us there. This involved our missing Turin, the scenery around which is well worth the photographer's attention, as I know of old, though the city itself has no feature of interest to the artist. At Milan we met with plenty of photographs of the Turin scenery, but, like the rest of the photographs to be seen in the shops as *souvenirs* of these places of a very inferior character.

At Milan, though there are several good portraitists, the views of the place and surroundings are of a very inferior quality. Shop after shop we

visited in the hope of getting a presentable view of the wondrous cathedral, but not one could I succeed in obtaining which would be worth having. In former days, no doubt, there was considerable difficulty in managing to get to a sufficient distance for a good point of view, and wide-angle lenses were not to be had. Now, however, that the Milan Improvements Company have cleared a broad space in front, this difficulty has been removed; but there seems no one who knows how to take advantage of it. Swing backs appear to be unknown, and every view is wretchedly distorted with converging perpendiculars—if such a phrase be admissible—owing to the tilting of the camera.

After spending a few days at Milan we started to visit my eldest son, resident in a small Lombard town off the main track of the railway, where he, as a civil engineer, was superintending the construction of some irrigation works. Here we stayed some days, seeing something of genuine Italian peasant or farmer life, which the ordinary tourist sees nothing of. A primitive life it is. The farmer or peasant grows his own hemp or flax, and steeps it in one of the numerous streams that are led through the place. It is then beaten or rubbed out with the hand, the women specially joining in this labour, and afterwards spinning it with the distaff. When spun it is woven in a primitive handloom into linen for household use; or it is made into ropes by means equally primitive in their arrangement. All these would form numerous interesting groups and bits for the photographer, and tourists would gladly purchase if they could get these *souvenirs* of the peasant in his picturesque costume engaged in these various occupations, standing at the door of the tumble-down cottage, or driving his peculiar cart.

The country is one vast flat, and it is only the people and their dwellings that afford any food for the photographer, but these are well worth recording. To do this, however, the artist must leave the beaten track, and he will need some enthusiasm for his work to make him put up with the wretched accommodation he will meet with in his country tour. We, indeed, were fortunate in not having to avail ourselves of this, staying in very comfortable quarters with our relative. The heat was excessive—so much so, that the navvies engaged on the canal construction worked all but naked, wearing nothing but a shirt.

After a few days' stay, much interested in all we saw, we rejoined the railway and sped on our way to Arona, steamed along the shore of Lago

Maggiore to Pallanza, whence we travelled along the great Simplon route to a village where another son resided, who is connected with the gold mines of that district. Here we took up our abode, making excursions up the Val Anasca to the foot of Monte Rosa, and visiting the various Italian lakes, passing from one place to another on their shores, sometimes in rowing boats, sometimes in steamers, and sometimes by carriage. In this way we passed through scenes of lake and mountain to which no photographer that we could meet with had done justice. At every place we stopped there were plenty of photographs offered to the tourist, from the humble *carte-de-visite* size to the more pretentious 12 by 10, but not one did I see that was worth having.

Landscape photography seems not to be understood by the Italians. They seem to set the camera down anywhere, without any consideration for picturesque effect—no judgment whatever exercised as to light and shade, and the result is a hard black and white picture with a clean white paper sky, wholly devoid of anything like artistic effect.

What would not a Wilson or an England effect here! I wish they who have done so much for Scotland and Switzerland could be persuaded to cross the Alps and give us the benefit of their labours in these charming scenes. Surely it would answer their purpose to spend some months in these parts. I can promise them, in addition to scenery of surpassing beauty and variety, good hotels and not expensive, with the means of locomotion easy and cheap. If the foregoing lines should induce them or some other competent artists to pay a visit to these regions, I shall, I hope, have done some service to photography and the diffusion of art.

P. LE NEVE FOSTER, M.A., Cantab.

THE EARS AND THUMBS OF CONTENTION.

WHO that has read the graphic account of the great diversity of shapes and characteristics of ears given by Mr. O. G. Rejlander in *THE BRITISH JOURNAL PHOTOGRAPHIC ALMANAC* will not feel an interest in this topic? Who, once more, that has read the evidence in the Tichborne trial, given in our number for December 22, by Colonel Stuart Wortley, Mr. Savage, and Mr. J. T. Taylor, will not feel some interest in thumbs? On both of these topics the Attorney-General enlarged at much length in his speech on Tuesday last.

Respecting the former subject—that of the ears—the Attorney-General said:—I come now to another member of the body upon which we have had a good deal of evidence. If the evidence you have got before you be true, the claimant and Roger Tichborne cannot be the same. That is my argument upon the question as to his ears. We have got unusually full and distinct proof about the ears, and I'll demonstrate to you that they cannot be the same on the ears alone. Now, the ears are not so unimportant as some people may be inclined to make out, although, I dare say, my learned friend will by-and-by treat the ears with summary contempt. First of all, M'Cann distinctly recognises him by his ears amongst other things. Then we find that Mr. Baigent in his affidavit said something upon the same subject:—"I then left with Mr. Holmes, and accompanied him to the Swan Hotel. On entering the large room facing the street I saw the plaintiff, whom I recognised instantly as the eldest son of the late Sir James Doughty Tichborne, whom I formerly so well knew. His eyes, the upper part of his head, and his ears were unmistakable, and his voice had quite an effect upon me, and was alone quite enough to convince me." At this time, when Mr. Baigent made his affidavit, he knew nothing about the Chili photographs. He probably knew nothing about either of them; certainly not of the one in the case produced here. These Chili daguerreotypes were in due time proved in the case to be good and true likenesses of the real Roger Tichborne, and we have Roger Tichborne's own letter upon the subject

describing how they were taken. He sent one, as you may remember, to Lady Doughty, and one to Sir James and Lady Tichborne. Finding that they were authentic and good likenesses, and knowing this at the time when he came to be examined (he himself having said in his affidavit that he recognised the plaintiff by his ears), and finding that the daguerreotypes would not suit the ears theory, Mr. Baigent, in his cross-examination, turns round upon the daguerreotypes and tries to discredit and throw dirt upon them, knowing full well, if they were to be depended upon, the ears of Roger Tichborne and this man could not be the ears of the same persons. He is acute enough to see that, and so when he is examined he turns round upon the daguerreotypes. He says it was a badly-formed ear; but Sir William Fergusson, on the other hand, said it was a large ear, but remarkably handsome. He was shown the photograph at this point, and asked whether the ear as it there appeared represented the ear of Roger Tichborne, and he said he thought not. The cross-examination goes on to say:—"The position in the photograph is not such as to give a good idea of the ear. Is this daguerreotype a fair representation of Roger Tichborne as you last saw him?—It is a very poor representation of him. I don't speak of it as a work of art, Mr. Baigent, but does it faithfully represent the features, to the best of your recollection?—I do not consider it a faithful or good portrait. Does it faithfully represent the features?—No, it does not give you a faithful representation of the man as he was. I cannot say, but it is something like him, but it is a very poor thing. Is there any mistaking it?—Oh, it is over-solarised, burnt out, and everything else. Is it a picture you could look at without knowing it was intended for Sir Roger Tichborne?—You must look at it two or three times before you can make it out. Does it represent the ear?—It does not convey an idea of the ear of Roger Tichborne." At this time Mr. Baigent knew perfectly well what we were going to say about the ears. At this stage of his cross-examination there was an adjournment, and when he comes back into the box he is questioned further about the other daguerreotype, and he then said neither of the pictures gave a representation of Roger Tichborne's ears. He throws dirt, you see, on the daguerreotypes, but he asserts they show the same sort of ear that the claimant now has. But my learned friend called Colonel Stuart Wortley. We are told he is a great authority in matters of photographs. It is no part of my duty to dispute that. I am not aware of it; but I have no doubt he knows more about photographs than Mr. Baigent. But Colonel Wortley

very likely did not know of this point about the ears. The Lord Chief Justice, in the course of the cross-examination, said:—"I should call your attention to this: that it has been suggested that the lobe of the ear might have extended below the junction, and that a different effect is produced by some defect of the light. Do you find indications of that?" Colonel Wortley, in reply to this question, said he did not think there was any such indication. We have, gentlemen, got thus far. The ear is a very important question. We have got Mr. Baigent first of all making importance of it in his affidavit, and we have got other witnesses who do the same. We have got it that Mr. Baigent at that time did not know of the existence of these Chili photographs, and that when he did know about them he said two things—first, that they were very bad, and not to be depended upon as likenesses; and next, that they showed the dependent ear of the claimant. Colonel Stuart Wortley is called on the same side as an authority, and he says they are to be depended upon, and do not show that upon which Mr. Baigent insists. That is so far as we have got in the great ear question, and by your leave we will here break off for our usual adjournment. [The adjournment here took place accordingly.] Let me say, gentlemen, I quite feel—and I am afraid I may have shown that I feel—the tedium of this part of the case; but if I am right it is the most important part of the case of all. This ought to be conclusive, because you may argue about mind and memory, but if the body be not the same there is practically an end of the case. Pray don't suppose I suggest that you are not the kindest of listeners. I do nothing of the kind. I am only apologising for appearing to take up time in apparently small matters, but if they are really made out to your satisfaction they are of enormous importance in the consideration of the case. If I show that the ears of the plaintiff cannot be those of the real Roger; if I afterwards prove, as I hope to be able to do by other features, an utter physical irreconcilability between the two persons, you will find that there is an end of the claimant's case. Well, as I have pointed out, Mr. Baigent, for reasons that I then explained, attempted to throw discredit on the daguerreotypes, though Colonel Wortley has distinctly accredited them. When that failed the next thing the plaintiff's advisers tried to do was to show that his ears had, in point of fact, altered. According to their present shape it must be admitted that they are not like the ears of Roger in 1854. The daguerreotypes prove this much, so they had to fall back on the theory of the ears having altered. The persons examined to sustain this point were Dr. Sutherland, Mr.

Canton, and Sir William Fergusson. I shall refer to Mr. Canton's evidence first. He is asked whether he noticed the ears at all, and he answered that he did. Both had this peculiarity—that they were unusually large and the lobes were pendent. The rest of the ears was well formed, and then—"Does any portion of the ear change with age?—The skin covering the cartilage becomes more firmly adherent, and the lobe of the ear in those who get lean in old age becomes less. Supposing a person gradually increased in size, what would be the result?—The lobe of the ear would be increased also if the increased size of the ear was due to fat. Would the ear become more pendent?—As it would become changed with fat the ear would increase in all directions except above. A Juror: Does the ear lay in fat in common with other parts of the body?—As fat increases in other parts of the body, so it would be in the lobes of the ear. In this case I saw fat cheeks, and expected to see fat lobes." Mr. Canton does not say whether the lobes would change in character, or whether they would become adherent or non-adherent; but he says that on that part of the lobe which is not cartilaginous fat would grow, as I suppose it would. I did not cross-examine this witness, because he did not say anything it was necessary to challenge, and because I had got all I wanted from Sir William Fergusson. He, therefore, left the matter very much where one's sense would leave it, and does not come to the point at all.

The Judge—He says his attention was only called to the claimant's ear the day before.

The Attorney-General (assenting)—But Sir William Fergusson, examined and cross-examined, goes much further in the matter:—"Is there anything you recollect as a peculiarity?—Very recently, about the lobes of the ear. What was that peculiarity?—The lobe was much larger than is the case with most people. The Lord Chief Justice: In what way larger?—Witness: It was broader and softer to the touch than is common. The Attorney-General: Do they depend much or adhere?—They depend so much that they strike one at once. When you say depend, what do you mean?—They hang down. Below the junction with the side of the face, do you mean?—Yes, they hang down. Are they free?—Yes, they are free. They are big lobes, handsome, and very conspicuous. They are, however, big, and many people would object to them on that account. Was there anything to show whether they were in a natural or unnatural condition?—Decidedly in a natural condition. Do you

think any pulling would have produced the condition in which you saw the lobes of the plaintiff's ears?—No; I think not.” Then in cross-examination by the jury, the witness was asked whether he thought a great increase in the bulk of the body generally would have affected the size of the ears, and what was Sir Wm. Fergusson's answer? He said, No; and that it was a rare thing for fat to be deposited on the ear—in fact, never; but we sometimes saw tumours in that locality, which he, however, explained were very different things from fat—so distinct, in fact, that it was impossible to mistake the one for the other. Sir William Fergusson, therefore, differs from Mr. Canton about the matter of fat on the ears. Mr. Canton says that fat is deposited on the ear where it is not cartilaginous and does not adhere to the cheek or side of the face; but Sir William Fergusson says, “No, it is a rare thing for fat to be deposited on the ear; in fact, never, I may say.” In effect, Sir W. Fergusson says there is no appearance of any trick or pulling the ears having been practised, and therefore the fair conclusion is that what he saw in 1871 he would have seen on the same ear in 1854, and there I must leave it. Dr. Sutherland is then called and examined about the ears:—“Was your attention called to his ears?—Yes. When?—On the 9th of November. Would you agree with Sir W. Fergusson that the plaintiff's ears are large, well-formed and unusually pendent?—Yes. Are they what you would call decidedly and distinctly detached ears?—Decidedly. Are there some ears which are perfectly lobeless?—No doubt. And others in which the lobe is detached?—Yes. Do you agree with Sir W. Fergusson that if a man has a lobeless ear it never becomes pendent?—I should think it would not. And anatomically you would say that if a man has the one kind of ear it never becomes the other?—Decidedly.” You thus perceive that Dr. Sutherland carries the matter a step further, and he says that there are for general purposes two classes of ears. His opinion is, he says, and also his experience, that an ear once attached never becomes detached and dependent. So far the scientific evidence. There are two medical gentlemen on our side—Mr. Seymour Haden, and Mr. Bernard Holt—who, as you will remember, saw the plaintiff when the rest of us saw him, and they will tell you that Dr. Sutherland is perfectly right in saying that a dependent ear never does come out of an attached one, and that Sir William Fergusson is right in saying that substantially the ear remains the same all through life. Take the Chili daguerreotypes and compare them with the photographs of the claimant. In the Chili daguerreotypes of the real Roger Tichborne,

which Col. Wortley says are for this purpose to be depended upon, you will find that the ears are distinctly adhering, but in all the photographs of the claimant where the ear is shown the ear is distinctly not adhering to the face. Take the uncontradicted evidence that the one class of ear never becomes the other class of ear, and draw for yourselves the conclusion. The ears of the real Roger were of the one kind, the ears of the claimant are of the other, and as it is not contradicted or disputed that the ears never change, it is plain that they are not the ears of the same man. Therefore, the two sets of ears belong to two different persons, and that is my case. Therefore, gentlemen, if you have no objection, I should like you to pause for a moment, and look at the photographs—first, the Chili photograph of Roger Tichborne, and then one of the photographs taken of the claimant since his return from Australia.

The Foreman—If you wish it, Mr. Attorney-General, of course we will do so with the greatest pleasure; but the point is so thoroughly before every one of us that we really do not think it is necessary. If you remember, the point was raised by us in the first instance, and it is most clearly before us now.

The Attorney-General—Very well, gentlemen. That is so, and I am very glad you see the point upon which I insist. There will therefore be no need for me to spend any more time about it, for we have plenty to do yet. So much, then, for the ears. The next point we come to is the hands in general. I have little to say about them, but then we come to a most important part of the question, one which my learned friend with great solemnity said more than once he thought was of the greatest importance—that is the question of the thumb. And I ask your attention to the way in which this has arisen, and the charge made in connection with it. It is the thumb of the left hand. You saw the plaintiff in the box. You saw him for a terribly long time—for many days. Did you ever see his left hand ungloved? Now, I ask you that question. Had he not always got a glove on his left hand? Did you ever see his left thumb? That the habit of the man was to sit with a glove on the left hand I am quite certain, and I believe he always did it. The case went on something like seventy days before we heard a word about the thumb, and the first we heard about it was in the re-examination of Mr. Baigent. It is remarkable the terms in which he introduced it, and the terms in which the charge is made, and the persons against whom it was made. “Attend to me

carefully,” says Mr. Serjeant Ballantine; “were the photographs shown taken from the daguerreotype?” “Yes, they were.” “Has the daguerreotype been in any way altered since it was taken?” and he answers, “Yes, certainly; it has been tampered with.” Then he is asked, “You say it has been tampered with; what has been done with it?” and his reply was, “The thumb and the greater part of the hand has been smudged out.” Now that is the first time in the case that anything was said about the thumb. The Lord Chief Justice then asked by whom the photograph was produced, when Mr. Serjeant Ballantine said, “You may take it that it has throughout been in the possession of the defendants.” The Lord Chief Justice remarked, “I always thought it was produced by the Court of Chancery.” The witness, however, said, “No, it was produced by Mr. Bowker.” The Lord Chief Justice then asked whether it was meant to be suggested that it had been altered since the adjournment, and my learned friend assents to that. Then Mr. Baigent, in answer to a question, says—“My belief is that it was not in the same state as it is now some few days after the first trial commenced last May”—so that there is no doubt that my learned friend and Mr. Baigent both meant to impute that this daguerreotype had been tampered with by the defendants since the trial began in May. Mr. Baigent further says—“I don’t pledge my oath to that. That is my impression. I did not notice it. I certainly should have noticed it I believe.” Then I say, “As I understood it, it may be taken as a fact that it comes from Mr. Bowker’s possession, and it should also be taken as a fact that from Mr. Bowker it was handed in, and has been in the possession of the Court ever since.” Then Mr. Serjeant Ballantine says—“*De die in diem* it has been in the possession of Mr. Bowker ever since.” Do not let us have any mistake about it. I don’t think my learned friend can say that I have not dealt fairly by him in calling your attention to this matter. Baigent makes a charge against the attorney in the cause of having tampered with this daguerreotype since the 6th of May. If English words mean anything, they mean that my learned friend says—“There have been suggestions against every human being in the case produced on the part of the plaintiff.” I do not know what he means by that. “I shall not shrink from any imputation.” Nobody has suggested it of my learned friend. Certainly not. “If there is any ground for a charge I will not hesitate to make it,” and then I say, with perhaps a superfluous interposition, “Of that I am quite certain.” Now, what is our answer? In the first place, that Mr. Bowker had himself multiplied copies from the unaltered daguerreotype—that he had

himself given the unaltered daguerreotype^[1] to Mr. Savage, at Winchester, who had multiplied photographs from it by scores and fifties, and that in every one of the positives from his negatives the thumb or hand remained exactly as it was in the daguerreotype from which it was originally struck off long before the beginning of the trial. Our first answer then was this:—"You accuse Mr. Bowker of having tampered with an original, the original state of which he had secured beyond all possibility of doubt by himself multiplying copies of it. No man in his senses could have been such a fool as to mutilate a thing the different state of which he had already put beyond all doubt and question." Our second answer was:—"That, from the time of the trial, or, at all events, from the time of the adjournment to the time it was said to be tampered with, Mr. Bowker had no control over it. There were two boxes, of which Mr. Bowker had the key of the small one, containing the daguerreotype, but that small box was enclosed in a larger one, of which Mr. Dobinson had another and a different key, and neither of these gentlemen could get at the daguerreotype without the consent of the other. The large box was never opened from the time it was locked up to the time it was produced here again, and therefore it is practically impossible that Mr. Bowker could have had anything to do with mutilating or interfering with it." Our answer, then, was complete and overwhelming—we did not know what was the cause of the disappearance of the thumb. That it was not in the same state in which it was when photographs were taken from it appeared by the production of the daguerreotypes and photographs; but when the change took place and how it took place we had not the least idea. We made inquiries and nobody could tell us. "We were not in full force at that time, because, as my learned friend Mr. Jeune had gone, on the part of the plaintiff, to Australia, so Mr. Purcell had also gone, on the part of the defendant, to Australia. About the time that the sittings broke up, Mr. Purcell returned from Australia, and among the consultations and discussions which ensued this matter of the daguerreotype turned up. "Oh!" says Mr. Purcell, "I am the person that did that, and they might have known it well." What happened was this. He went out to execute the Chili Commission, and as one of the things by which to test the Chili witnesses, he took out this very daguerreotype of the real Roger for the purpose of asking them if it was like the man they recollected, and who had been writing to them, and they all of them said they did not recognise it. As part of his luggage he put it in a box that was too big for it. In fact it was not

properly secured. It was fixed at the back of the frame by little bits of metal, which were attached to and formed part of the frame, and were bent down—little projecting saw-teeth, which were bent down over the daguerreotype, and held it tightly in the frame. This daguerreotype was placed in a little box, not properly secured, and the little box was in a portion of the luggage; and on the journey from England to Valparaiso, or from Valparaiso to Melipilla, the daguerreotype got out of its case. It got knocked about, and one or two of these little teeth becoming unfastened the daguerreotype was unfastened.^[2] When Mr. Purcell arrived at Melipilla he took it out and found it in this state. He then at once went to a photographer and had it refastened down. It was used in the Chili Commission in its present state, and could not have been injured in the Commission. Mr. Purcell will tell you how it took place, and this daguerreotype has never been used for the purposes of photography since its return from Chili. The piece of paper at the back, Mr. Purcell will tell you, was pasted on by himself, and here are the signatures of the Chili Commissioners on the paper as written at Chili, and since then the daguerreotype could not possibly have been interfered with. Therefore there can be no mystery about it, and yet in the face of what I mentioned to you before, that Mr. Bowker had himself photographed the uninjured daguerreotype by scores, and in the face of the other facts of the case, my learned friend, Mr. Sergeant Ballantine, allowed his case to come to an end with that offensive imputation against the advisers of the defendants, and never explained or withdrew it. Well, is the thumb in the photograph like the plaintiff's thumb? Nobody ever said it was from the beginning to the end of the case but Mr. Baigent.^[3] This point is due entirely to the ingenuity of that gentleman. His attention appears to have been drawn to the condition of the daguerreotype early in May. No one's attention was drawn to the thumb till late in November, and the first word that is said about it is said by Mr. Baigent in re-examination. Now, you know that in May, just about the time that the attention, as I have said, of Mr. Baigent appears, by his examination-in-chief, to have been drawn to the photograph, I was asking for the inspection of the plaintiff's person for any marks on him upon which it was intended hereafter to place reliance. My learned friend objected, for the reasons you know, to that proposal. In his own interests he refused to allow that examination to take place, and he must not now complain at the remarks which all the circumstances give rise to in my mind. I don't for a moment say—and when I say a thing I trust you have learned I mean it—I

don't for a moment say that my learned friend had any reason to think there was any peculiarity about the thumb when he refused inspection. But somebody knew, and that somebody instructed my learned friend to refuse the examination. I point out to you, once more, that no human being ever said a word about the thumb till Mr. Baigent mentioned it. Neither Baigent, nor Hopkins, nor the plaintiff himself in his letter to his mother in which he mentioned the brown mark ever suggested the existence of a peculiarity in the thumb. My learned friend hears of it for the first time only a few days before the end of his case, on the 12th of December. That is a remarkable thing. My learned friend more than once or twice says it is highly important. Sir William Fergusson, when he first examined the plaintiff, did not notice any peculiarity in the thumb, but on the day he appeared in the box he said he made an examination of the thumb of the left hand. He noticed there was a peculiarity in the thumb. It was peculiarly pointed at the extremity. The fleshy part projected beyond the nail more than usual, and more than on the other hand. The nail seemed to be perfectly developed, and it struck him that the plaintiff had been in the habit of biting his nails, which he afterwards learned was the case. Being shown the photograph, he said it was an exact representation of the claimant. Then in cross-examination he says—"I don't think such a thumb could be made by manipulation. I don't think any pressure would produce it. The nail is smaller on this thumb than on the other. The whole thumb is smaller at the point. The condition, I don't think, would be formed or made more marked by biting the nails. I never had my attention called to it until yesterday morning."

The Attorney-General is at fault here; for Mr. Savage, as reported in our issue of the 22nd December, page 603, distinctly said that his negative was made from a *print* that had previously been made from the daguerreotype, and not from the daguerreotype itself.—EDS.

Photographers will smile at the idea of the serrated edge of a brass mount which has been clamped over the *back* of a plate getting inside of the picture, between the glass and the surface.—EDS.

According to the report of the case that appeared in our pages the photographic witnesses examined in the case have also asserted this.—EDS.

The Lord Chief Justice—I don't understand that Sir W. Fergusson's evidence was so much directed to the flesh protruding beyond the nail as to the peculiarity in the shape of the thumb. That was the peculiarity which he meant could not be produced.

The Attorney-General—It comes to this: Sir William Fergusson does not think that biting would do it, nor that it could be done artificially. But the plaintiff had not been examined by our medical men then, and I was not in a condition to cross-examine Sir W. Fergusson. Mr. Canton is also examined, and states that his attention was called to the thumb for the first time the day before. There was this peculiarity about it—that the nail was attached to the part beneath to a less extent than it was in the case of the nail on the other thumb. On being asked if such a condition of the left thumb could have been produced by any known process which a man could exercise on himself, Mr. Canton said that the only thing he could conceive would be a person continually biting his nails. It really comes to this, that if the man had let his nail alone, the difference between the two nails would be greatly less. That is the evidence of a witness for the plaintiff. He is of opinion that the nail has been subjected to treatment for the purpose of producing a difference, and that if it had not been there would have been much less difference. Now, what does that show? That the nail has been manipulated, prepared, gone through a process of concoction, and not a word about it was mentioned to any human being, however closely connected or intimate with the plaintiff, until the 13th December, when the thumb had been put in a proper state for examination. When the doctors examined the plaintiff in November, their attention was not called to the thumb. I suppose it was not then in a state to be examined; but in December they were suddenly called out of court, and in some back room of this Sessions-house this thumb was shown to them. Then this astounding charge was made of our having tampered with the daguerreotype for the purpose of destroying evidence which we ourselves had actually preserved. As far as I understand the evidence of the medical men it will be this. They will express a clear and confident opinion, from a scientific examination of the thumb, that it is a non-natural thumb, by which I mean that it presents appearances which show that it has not been allowed to take its natural course, but has been manipulated and treated by the plaintiff or some one in his interest—manufactured, concocted, and got up—for the purpose of this examination, and now we can understand why the glove was always kept on the left hand, and why not a single syllable was said about it until it was ready for inspection. You can also understand why the secret was only committed to Baigent. It was not even communicated to Miss Braine. My learned friend, Mr. Serjeant Ballantine, said it was an important matter, and indeed it is.

Mr. Baigent himself did not venture to say a word about the thumb until the pressure of cross-examination was removed; he reserved it for his re-examination, when the mouth of my learned friend, Mr. Hawkins, was closed. Yet they searched all over the man—they drew attention to wretched little scratches on his ankles, which you could hardly see, and to a little white mark on the middle of his eyelid, which I confess I could not see—these they hunted out for demonstration; but this patent thing, which he could not put his hand on a piece of paper to steady it while he was writing, he could not wash his hands or use a knife and fork at dinner without exhibiting—this most staring and glaring peculiarity was never mentioned to any human being until the 13th December. In any other but the Tichborne case what would have been thought of this? Take a railway accident case. What would a jury say if alleged injury were brought before them at the last hour which neither the plaintiff nor his counsel had ever said a word about before? I know what would have been said of myself and Mr. Hawkins if we had been counsel for the plaintiff. But because this is the Tichborne case it is allowed, and my learned friend, Mr. Serjeant Ballantine, with an air of great solemnity, says—“This is a very important matter, and a great deal more will be heard of it.” I understand my doctors will say the nail does not adhere so closely to the under surface as the nail does on the other hand—that it is perfectly easy to disconnect them—that it may be done by perseverance, by introducing little pins and things of that kind, and by gradually and steadily working them to disconnect the under surface of the nail from the under surface of the flesh, and that in this way you can arrest the “formative growth” of the nail, and they will tell you that that has actually been done. You will look at the photographs for yourselves, and I ask you to come to the conclusion that there is no reasonable pretence for saying that Roger Tichborne had such a thumb. There is no one to suggest that Roger Tichborne had a peculiar thumb—no one till Mr. Baigent suggests it on his re-examination; and I submit that the whole affair has been trumped up to produce an impression when they found that the other marks failed them.

Meetings of Societies.

MEETINGS OF SOCIETIES FOR NEXT WEEK.

Date of Meeting.	Name of Society.	Place of Meeting.
Feb. 7th	Edinburgh	Hall, 5, St. Andrew-square.
Feb. 8th	South London	Arundel Hall, Arundel-street.
Feb. 8th	Manchester	Memorial Hall, Albert-square.

EDINBURGH PHOTOGRAPHIC SOCIETY.

The second popular meeting of the season was held in Queen-street Hall on the evening of Wednesday, the 24th ult. Every available place in the hall was filled.

The exhibition consisted of a very fine series of views in Ireland, kindly lent by Messrs. Pumphrey Brothers, of Birmingham, and of a number of copies of Erskine Nicol's pictures of Irish character, prepared from negatives kindly lent for the purpose by Mr. Tunny. The pictures were arranged in the form of a tour, beginning in the north at the Giant's Causeway, working south to Dublin, and including the Lakes of Killarney and some of the finest scenery in the country.

The descriptive lecture was undertaken by Mr. J. Richard Marquis, R.H.A.; but being evidently new to the business, and unaccustomed to speak in the dark, the lecturer failed to come up to the expectations of the audience. We hope, however, to hear Mr. Marquis again, as he has in him the elements from which good lecturers are formed; and from his ability as an artist, and his thorough acquaintance with what constitutes a good picture, we are certain that with a little practice he will make a capital guide in such exhibitions.

The exhibition was thoroughly successful, and at its close hearty votes of thanks were awarded to Messrs. Pumphrey Brothers, Mr. Tunny, and the lecturer.

We may mention in connection with this that last evening (Thursday) the same pictures were exhibited as one of the series of popular lectures given under the auspices of the Northern District Lecture Association. On this occasion the lecture was delivered by Dr. John Nicol, who stated at the beginning that he was in a position to do what rarely could practically be done by a lecturer—that is, praise his own lecture. He (Dr. Nicol), however, went on to say:—"The lecture is not mine, but has been kindly sent along with the pictures by Messrs. Pumphrey; and it is so much better than anything that I could produce, I shall give it just as it is, except, perhaps, the introduction here and there of a story or anecdote, by way of a 'mouth-

opener,' as our transatlantic friends would say." Both lecture and exhibition were very successful, and elicited almost continuous applause.

At the close Dr. Nicol exhibited the recently-patented "wheel of life," as manufactured by Messrs. Pumphrey, and described in THE BRITISH JOURNAL OF PHOTOGRAPHY a few weeks since. He explained the principle of its construction, and gave much credit to the inventor, who, he said, had succeeded in doing admirably what he (Dr. Nicol) and many others had vainly tried to do for years.

At the termination of the exhibition, votes of thanks were awarded to Dr. Nicol, to Messrs. Pumphrey for the pictures, and to Messrs. Yerbury and Lothian for the admirable way in which they had managed the lantern.

Correspondence.

THE CONSTITUTION OF MATTER AND ITS MOTIONS.—UNIVERSAL GRAVITATION
PRODUCED BY THE MOTIONS OF ETHER.—HYPOTHESIS OF M. L'ABBÉ LERAY.
—M. RICHE ON THE NEW CHEMICAL NOTATION.—NEW SENSITIVE PIGMENT
PAPER.

In one of my recent letters to this Journal I was deploring the want of some grand hypothesis which would embrace in one general law all the mechanical phenomena of nature—universal gravitation, cohesion, chemical affinity, electricity, magnetism, light, heat, &c.—and lo! just such a hypothesis has turned up and been explained to me by its author—a man whom I am compelled to regard, without any exaggeration, as one of the great geniuses of the present day. I allude to M. L'Abbé Leray, author of a work to which I referred in a recent letter, entitled *Constitution de la Matière et ses Mouvements*.

I will endeavour to give an outline—but it can only be an outline—of this startling hypothesis, which makes universal gravitation an effect of the motions of ether, and dispels the idea hitherto entertained that it is an attractive force inherent in particles of matter, by which they are enabled to exert a pull upon other particles at a distance from them in space.

Sir Isaac Newton, the immortal discoverer of the law of universal gravitation (which asserts that every particle of matter in the universe attracts every other particle with a force which varies directly as the mass and inversely as the square of the distance), himself believed that this law would be found, at some future time, to result from the motions of a subtle fluid which occupied space, the atoms of which, by impinging against ponderable matter, produce the observed effect of weight or gravitation. He could not conceive of ponderable matter possessing any *inherent* property by which it was enabled to act upon other ponderable matter *at a distance*. He could not really believe in an inherent central force of attraction residing

in a material atom, by which it could draw towards itself another such atom situated at a distance, or that it could affect other matter in any way than by the impact of intervening atoms; and he predicted that some day the *cause* of universal gravitation would be discovered. That day seems now to have arrived, and the discovery to have been actually made.

I had read M. Leray's extraordinary work with a great deal of interest, but there were still some unexplained points in his hypothesis which remained to be cleared up and some difficulties to be discussed. As he resides at an ecclesiastical institution within an easy walk of Redon, I visited him a few days ago, by appointment, for the express purpose of having a long talk over his hypothesis; and a very pleasant afternoon I had with him, leaving more deeply impressed than ever with a sense of his great abilities and originality. In fact, I seem to have made the acquaintance of a second Sir Isaac or Laplace.

The following is an outline of his theory; but of course no demonstration of any part of it can be made within the limits of this letter:—

Space is filled with a subtle ether, consisting of atoms in motion. These atoms are elastic—a property which they possess in virtue of being able to change their form, though not their volume, during impact and to recover it again. Their form is spherical, they are all equal, and their diameter is very small compared with that of the atoms of ponderable matter, and also with their general distance apart. This ether is, therefore, an exceedingly rare medium. When the atoms impinge against each other they rebound like billiard balls, and in all their motions they obey the common mechanical laws of inertia and impact, and no other laws whatever. They cannot act upon each other at a distance, and therefore no attractive or repulsive force exists between them.

If only one of these atoms were to exist in space it would move in a straight line with uniform velocity until it reached the limit of space; that is to say, the boundary by which creation is limited—the boundary which separates entity from nonentity. Here, being elastic, it would be reflected, and would then follow another rectilinear course until it again encountered in another point the boundary of space, where it would be again reflected; and so on for ever!

If we imagine space filled with an enormous number of such atoms it will follow that *at every point in space there will be small parallel currents of them moving in all directions*. Their distance apart being great in proportion to their size, two contrary currents will not annihilate each other, but by far the greater number of atoms in one current will pass those in the other current without impact. Those atoms which do impinge against opposite atoms, at various angles of incidence, will rebound and join other currents which are moving in their new direction.

The state of things above described constitutes what is called “mobile equilibrium;” for what one current loses by meeting another in an opposite direction will be imparted to surrounding currents, and these, in their turn, will give back equal to what they have acquired, so that compensation will be made, and thus the laws of conservation of force, and of *vis viva*, will be satisfied.

The velocity with which the atoms move is enormous, and millions of times greater than the velocity of light.

The reader will observe that there is a vast difference between the mobile equilibrium of this ether and the equilibrium of air or gas confined in a closed vessel. The reason why particles of a gas appear to repel each other is because the ethereal undulations of heat are vibrating between them. By reducing the temperature and increasing the pressure gases may be liquified or solidified, in which states no repulsion exists between their particles.

All ponderable matter is porous; its ultimate atoms are spheres much larger than the atoms of ether, and much farther apart; the currents of ether can, therefore, pass through a ponderable body in all directions.

When a current of ether passes through a ponderable body some of the atoms of ether strike the atoms of the body and rebound; the current, after passing through the body, will, therefore, be weakened, according to the number of its atoms which have rebounded in altered directions—that is to say, according to the number of atoms of the body which have been struck by atoms of ether. The greater the mass of a body the greater will be the weakening of the currents of ether which have passed through it. A current of ether weakened by passing through a body will gradually regain its original strength by passing through space, since it will be continually reinforced by other atoms moving in the same direction as itself.

A single ponderable atom in the midst of currents of ether will be in equilibrium under their action, because it will be struck equally in all directions.

But the atoms of a ponderable body will be put into vibratory motion by the passage through it of currents of ether; these internal motions may enable us to account for light, heat, magnetism, &c.

When two ponderable bodies exist in space, and currents of ether pass through them, the two bodies will be impelled towards each other, because the currents of ether that are between them and tend to keep them apart are weakened by having passed through the bodies, and are, therefore, weaker than the currents which impel them towards each other. This explains what has been called the “attractive force of gravity.”

It will be observed that since currents of ether pass through the bodies in all directions, the weakened currents between the bodies will be included within a sort of conical space. The law of attraction according to the inverse square of the distance is thus accounted for.

Since the weight of a body is the same, no matter how it is turned about, it follows that the ultimate atoms of all ponderable matter must be spherical. It follows also from the hypothesis that all the spherical atoms of ponderable matter are equal, and that there is, chemically speaking, but one simple substance—the apparent variety depending upon the mode of aggregation of the atoms into molecules.

Crystals are formed by arranging these spheres in the same way as you may arrange marbles or pile up cannon balls.

There is nothing in the hypothesis to interfere with the undulating theory of light, or with any theory that reposes strictly upon observed facts; but this we may discuss on a future occasion.

I need hardly say that it is in consequence of their great velocity that the atoms of ether acquire sufficient momentum to communicate sensible motion to ponderable matter.

Ponderable matter may possibly be composed of the aggregation of ethereal atoms; but M. Leray thinks not. He can see no good reason why it should be so.

Cohesion and chemical affinity may be explained on this hypothesis. Its leading feature is that it explains how such natural phenomena as do not involve vital or mental action may be explained on the simplest mechanical principles, and without involving that “bugaboo,” *action at a distance*. Of course, Dr. Frankland’s ideas of “bonds,” “active and latent atomicities,” &c., are inadmissible on this hypothesis.

The demonstrations are rigorously given, and the work involves a good deal of high mathematics. It is utterly impossible to do justice to the theory in the above brief sketch of it. The theorems of Euclid, if thus stated, would many of them appear improbable and absurd. The work itself can be procured from M. Gauthier Villars, 55, Quai des Grands Augustins, Paris, price three francs. It is copiously illustrated with woodcuts. A new edition has just appeared.

Some idea of the distance between the atoms of ponderable matter, when in the form of gas, may be gathered from a remark of Dr. Mann’s in his *Guide to a Knowledge of Life*, at page 13, where he says:—

“It can be shown to be highly probable that the ultimate atoms of gases are at least one hundred times their own diameter asunder even when those gases are held in confined vessels.”

The earth and the moon are, therefore, about three times as near together, in proportion to the diameter of the earth, as two atoms of a gas are, if the above remark be true.

In Sir Isaac Newton’s corpuscular theory of light the atoms emitted from the sun were supposed to follow one another at a distance of about a thousand miles apart! Under such circumstances the impact of two atoms of intersecting rays of light would be a comparatively rare event.

M. Leraf asserts that the law of gravitation is only an approximation to the truth, and that it is modified by the volumes of bodies. The proof of this he expects will be found some day in the motions of comets, which rapidly change their volume.

Elective affinity he supposes to depend upon the different forms of crystals, two crystals which present plane faces towards each other being more easily pushed together by the atoms of ether than two crystals in

which a solid angle or an edge of one is presented to a plane face of the other.

The sun, planets, fixed stars, nebulae, &c., are, of course, perpetually riddled through and through, in all directions, by currents of ether. That is why the heavenly bodies gravitate towards each other, as explained in a preceding paragraph.

With respect to reflection at the boundary of space, it is an idea which grows upon you the more you think of it. Enormous as creation is it is impossible to conceive of its having *no* limit. What, then, is beyond that limit?—Nothing. Not even space in which matter can exist; no *place* even for matter. On reaching the boundary which separates an entity (for space is an entity) from a nonentity matter must be reflected, if elastic; or it must roll for ever against the boundary of space, if inelastic. This conclusion seems to me inevitable; there is no escape from it.

In the new edition of M. Leray's book he modifies the theory which I have endeavoured briefly to explain in the foregoing paragraphs by supposing that, instead of one ether, there are two in a state of mixture, the second being a grosser fluid, and its atoms larger than those of the other. It is these larger atoms of the grosser fluid which, by their transversal vibrations, produce the phenomena of light, heat, &c. These larger atoms do not suffer the same swift motion of translation through space as the smaller atoms of the subtler fluid. They have no greater motion of translation than ponderable atoms have.

It may be asked—What is the difference between ponderable and imponderable matter, and why are the atoms of ether imponderable? To this query a satisfactory answer is given; but I must refer the reader to the book for it. Were I to enter upon any demonstrations an entire number of this Journal would not contain half that could be said.

I have proved in an independent manner, and different from that of Père Leray, that two equal, penetrable spheres of ponderable matter, existing in space at a distance apart which is large in proportion to their diameters, will be impelled towards each other by the impact of ethereal matters, according to a law which is approximately that of the inverse square of the distance. When the spheres are brought to within a much shorter distance of each other the law ceases to be approximately true. The law of gravitation may,

therefore, be only approximately true for particles of matter at a great distance apart in proportion to their diameters. The only observations which appear to confirm the law are those which have been made upon the heavenly bodies; and here we have a case of a distance apart many times the diameters of the bodies, even between satellites and their primaries.

But before any one can seriously accept this new hypothesis a vast deal more thought and study must be bestowed upon it than I have yet had time to give it.

I will send the demonstration referred to for insertion in a future number of this Journal if our Editors think fit. The subject is not foreign to photography, but intimately connected with it as a science.

According to the new hypothesis, new definitions must be given of **MASS** and **DENSITY**. According to M. Leray, “the mass of an atom is equal to its volume, and the mass of a body is equal to the sum of the volumes of its atoms.”

“If we call M the mass of a body, and V its apparent volume, the fraction M/V is the absolute density of the body. The absolute density is, therefore, unity for an atom, and varies from 0 to 1 for all bodies.”

If two bodies have the same apparent volume, their densities are proportional to their masses.

I have been looking through a capital French work on Chemistry, published in 1870, by M. Alfred Riche, lecturer at the Polytechnic School at Paris. He uses the old notation and table of equivalents; but strongly advises a change to the new, which he explains very nicely, and pretty much as our lecturer has done. Whenever the atomic weight of an element is given according to the new table its symbol has a bar drawn across it. Something of this sort should always be observed, in order to avoid confusion between the old and new formulæ.

I have just received a letter from Mr. J. R. Johnson, containing a most beautiful carbon print. He asks me what I think of it. My reply is simply this—that it is the most wonderfully fine print I have ever seen upon paper.

THOMAS SUTTON, B.A.

Redon, January 26, 1872.

THE RECENT SOLAR ECLIPSE.—NEW PHOTOGRAPHIC VENTURE AT THE
ANTIPODES.—PHOTOGRAPHY IN THE BUSH.

The most interesting event, in a photographic point of view, which I have to report is the departure of the scientific expedition to observe the total eclipse of the sun on the 12th of this month. The place selected for the observations is Cape Sidmouth, some three hundred miles south of Cape York, in Northern Australia.

The expedition has been organised by the Royal Society of Victoria, and the expense is met by private subscriptions, largely aided by grants from the several Colonial Governments. The Queensland Government steamer was also placed at the disposal of the party free of charge.

The various instruments necessary for the observations were sent from England by the Royal Society; but, owing to my absence from Sydney at the time the expedition sailed, I am unable to give any details of the arrangements for taking photographs of the eclipse. I hope, however, soon to be able to give a full description of the results.

The party consists of more than thirty gentlemen, the different branches of science being well represented; for botanists and geologists are taking advantage of the trip to make investigations in their own departments. For the astronomical observations Victoria sends her Government Astronomer, Mr. Ellery, at the head of a large staff of observers, and a photographer, Mr. Walters; while New South Wales sends Mr. Russell, the present, and the Rev. W. Scott, the late, Government Astronomer, and Mr. Merlin, of the "American and Australasian Photographic Company." From this double staff we may expect a large number of photographs and other valuable results.

The steamer left Sydney on the 27th of November, and will return by Christmas, before which time no news will reach us of the doings of the party. We shall, therefore, look forward to their return with much interest.

Professional photographers here do not now devote their attention so exclusively to portraiture as formerly. The "American and Australasian

Photographic Company” has announced its intention of photographing every house in the Australasian colonies! I suppose it finds it a profitable speculation, as it has already photographed a considerable number of towns, house by house. The day for each place is previously advertised, so that the inhabitants may put themselves and their dwellings in holiday attire. The photographs are to some extent used as advertisements.

I lately came across a photographer in the far interior, some 500 or 600 miles from Sydney. He had already travelled a still greater distance from Adelaide, in South Australia, from whence he had started on his tour. He was plying his vocation at the various sheep and cattle stations, and was apparently well patronised. I saw several of his groups of aboriginals, which were very good. The black fellows were highly delighted with their portraits, and were very anxious that copies should be sent to their friends in other districts.

Sydney, December 1, 1871. E. B. DOCKER, M.A.

P.S.—The unfortunate wreck of the mail steamer has deprived us of the journals for this month.—E. B. D.

CARBOLIC ACID.

To the EDITORS.

GENTLEMEN,—My attention has been called to an article in your issue of the 19th January, under the head “Correspondence,” by Mr. Thomas Sutton, containing several statements with reference to carbolic acid which it would be wrong to allow to remain uncontradicted.

First: he says carbolic acid is by no means a good antiseptic, and is very poisonous, and then refers to persons being lately poisoned by its *fumes* at Wolverhampton.

As to its poisonous nature: It is, of course, a poison if taken internally in quantity, but is not a virulent one taken in any reasonable or probable quantity. It is, perhaps, not out of place to say that if it should be taken internally in a concentrated form, by misadventure, large doses of castor and sweet oil immediately administered will materially counteract the poisonous effect of the acid. The fumes of the acid are perfectly harmless and may be breathed with impunity.

Mr. Sutton is labouring under a false impression with regard to the case to which he alludes at Wolverhampton, of which the following is the correct account:—Two dogs (not human beings) were *supposed* to have died from inhaling the fumes of carbolic acid emanating from a disinfecting powder sprinkled over the floor of a workshop in Wolverhampton, and, the following is an extract from the report of the chemist who examined one of the dogs:—“The disinfecting powder was not a carbolic acid powder, but an imitation; for it contained nothing but lime impregnated with tar, and was entirely innocent of any harm to the animals. Strychnine, however, was discovered in considerable quantity in most parts of the viscera and in the blood. I calculated that at least one grain and a-quarter of this poisonous alkaloid had been administered to the animal by some evil-disposed person or persons unknown.” Thus much for the poisoning of human beings lately at Wolverhampton by the fumes of carbolic acid.^[4]

The cattle show at the Agricultural Hall, Islington, has for the two last years been successfully disinfected and kept sweet with carbolic acid.

Had Mr. Sutton ever seen carbolic acid fumigation, or read about carbolic acid, he would not have made an assertion so utterly groundless. The writer has himself many times been for two or three days together in a room containing a large excess of carbolic acid fumes without experiencing any injurious effects.

With regard to the action of carbolic acid on the teeth, if Mr. Sutton will refer to an article in the *British Journal of Dental Science* for March, 1871, he will find "it is a powerful antiseptic, and invaluable for the arrest of any decay or decomposition of the teeth."

Mr. Sutton has quoted from a long letter of Dr. Dougall's to the *Lancet*, and I cannot do better than refer him to Dr. Sansom's able reply to it in the *Lancet* of January 13th. Dr. Sansom says that the white-cloud appearance in albuminous solutions to which carbolic acid has been added is often really no albuminous precipitate at all, but is caused by refractile globules of carbolic acid in a fine state of sub-division; also, that it has been shown that albuminous solutions are antisepted when carbolic acid exists in them in too feeble a proportion to cause any precipitate whatever. If carbolic acid acted as an antiseptic by coagulating albumen, agents which had a greater coagulating power would, *a fortiori*, be more powerful antiseptics, which has abundantly been proved not to be the case, and therefore the antiseptic properties of carbolic acid do not result solely from its power of coagulating albumen.

With respect to the assertion that the amount of carbolic acid vapour which could be tolerated in the air of a hospital ward would be entirely inadequate to act as a disinfectant, Dr. Sansom says his experiments have shown him that *carbolicised atmospheres* are efficient in preventing putrefaction and the growth of mouldiness, and more so than atmospheres impregnated with chloride of lime or sulphurous acid.

Dr. Sansom objects to the experiments recorded by Dr. Dougall, since tar oil (a crude product weak in carbolic acid, and possessing little or no volatile disinfectant constituent), and McDougall's powder (a mixture of sulphites of lime and magnesia with tar oil) were used.

As to carbolic acid not being a good antiseptic, the following reports, I think, fully prove the contrary:—

The late Dr. W. Allen Miller, F.R.S., preserved urine and fresh blood for three months by the simple addition of five per cent. of Calvert's carbolic disinfecting powder—a product containing fifteen per cent. of carbolic acid in a free state.

Mr. Wm. Crookes, F.R.S., says that he took some albumen from fresh eggs and mixed it with an equal bulk of water. By itself it became bad after nine days, and at the end of three weeks it smelt very strongly. He added to four bottles of the fluid respectively 1, 2½, 5, and 10 per cent. of carbolic acid powder (equivalent to 3/20, ¾, ¾, and 1½ per cent. of free carbolic acid). All kept good at ordinary temperatures for forty days. Blood with 1/15 per cent. of carbolic acid remained good for a month. Solutions of size, glue, and gum mixed with 1/15 per cent. of carbolic acid have remained for two months without becoming sour. Fresh yeast was washed with water containing one-tenth per cent. Its power of inducing fermentation was entirely destroyed.

Dr. F. Crace Calvert, F.R.S., in his paper on comparative disinfectants, gives the following results with antiseptics upon solutions of albumen:—

Antiseptic employed.	Percentage of antiseptic used.	Time in which acquired an offensive odour. Temperature 70° to 80° F.
Chloride of lime	5	16 days.
Tar oil	2	11 days.
Carbolic acid	2	remained sound six months

None

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5 days.

The writer preserved meat for ninety days, during a hot summer, by placing twelve ounces of fresh meat in a bottle containing one pound of water and five grains of carbolic acid. The mouth of the bottle was left open, and no offensive smell was emitted till the ninety-third day. The meat was, of course, unfit for food, and was merely experimented with to test the antiseptic power of carbolic acid.

The following is an extract from a report in *Compte Rendus de l'Académie des Sciences* of March 6th, 1871, by Messrs. Nelaton, Langier, and Payen, on experiments made at the Paris Morgue by M. Devergie:—

“During the heat of summer, when putrefying corpses in the Morgue continually emit a quantity of noxious gases that cannot be removed by ventilation or destroyed by chlorine or bleaching powder, we decided to prevent their production by trying to destroy the vitality of the germs of putrefaction, and thus prevent decomposition itself. We effected this by dissolving one litre of carbolic acid in 1,900 litres of water and irrigating the bodies with the solution thus made. Putrefaction was completely stopped, and disinfection was even obtained after reducing the quantity of acid by one-half. M. Devergie points out that water containing one four-thousandth part of carbolic acid proved sufficient during the intense heat of last summer to disinfect the deadhouse, without the aid of any shaft, when six or seven dead bodies were lying there. * * * * * Carbolic acid seems well adapted for the disinfection of rooms which have been occupied by persons suffering from infectious diseases; therefore, we recommend its use, after being dissolved in thirty times its weight of water, by sprinkling it on the floors, pavements, and staircases during the stay of patients in rooms and for a few days after their departure.”

According to Dr. Sansom carbolic acid is readily taken up by air, so that 159.44 cubic inches of air, at 60° Fah., contain one grain of carbolic acid. Air thus *carbolised* (currents excluded) entirely annuls putrefaction and fungoid manifestation on the surface of putrescible fluids, and such carbolised air is more permanently efficacious than air charged with the fumes of chloride of lime or sulphurous acid, and it may be breathed with impunity by mammals.

These few observations will, I think, satisfy your readers that Mr. Sutton's remarks are erroneous and without foundation. I shall be glad to learn that photographers have tried carbolic acid in the preservative solution for dry plates, and would recommend them to make a solution of carbolic acid, one part to one thousand parts of water, and then add their albumen to this solution to the strength they require it. Above all things it is essential that the carbolic acid be of good quality for photographic purposes; and I would recommend them to use an acid such as Calvert's No. 1 (gilt label) carbolic acid.—I am, yours, &c.,

REGINALD LE NEVE FOSTER.

Bradford, near Manchester, January 27, 1872.

THE TEST FOR ALBUMEN.

To the EDITORS.

GENTLEMEN,—It is over two years since I devised the carbolic mixture for detecting traces of albumen. At the time I did not think it of sufficient photographic interest to occupy your space in detailing experiments on the subject; but, as you do not remember the author's name, I beg to remind you that I am the author. It was published, I believe, as a note by a friend of mine—an eminent chemist and toxicologist—in a medical work.

On one occasion we were talking over the means of detecting albumen, and having experimented with phenol or “carbolic acid” for about eight years, many times in connection with albumen, I knew its properties well, and, on my attention being directed to a test for albumen, I commenced experiments with a mixture of phenol and acetic acid, then with the addition of alcohol, and finally with phenol and alcohol, equal parts by weight. My friend and self then went through a comparative set of experiments side by side—my friend taking his old nitric acid test and I my new phenol mixture—the result being that my test indicated albumen both in plain water and urine, diluted one in ten after the nitric acid failed to indicate any further.—I am, yours, &c.,

F. W. HART.

8, Kingsland Green, January 29, 1872.

SUGGESTION FOR A PHOTOGRAPHIC EXHIBITION.

To the EDITORS.

GENTLEMEN,—The continued complaints which one heard from people after the close of our photographic exhibition, that they had not seen it, did not know when it opened or when it had closed, and the strong interest they felt in it, induce me to believe that a comprehensive exhibition of photography in all its shapes—the new processes, the landscape and humanity of different countries, &c., &c.—might be made of very great interest, and to pay its way as well.

In the spring, and even now, London is filled with collections of paintings, which make photographs look tame. In the International Exhibition they are equally put out; and if they are to be seen and judged properly they must be in an exhibition by themselves. No intelligent collector hangs works in colour with photographs or engravings; and exhibition goers, passing from a gallery of pictures, will not stop to look at photographs. If, therefore, we could, under the direction of the photographic societies, make in October a collection of cosmopolitan photography, and connect with it a display and comparison of lenses of all makers, we should enable the art to claim its just consideration.

Colonel Stuart Wortley's suggestion as to contributions of negatives would give a photographic exhibition, if adopted, a special technical interest; and we might in this include examples of negatives by the dry processes.

The question of retouching might be decided without difference of opinion by having two classes of works—one touched on the negative, and the other in which no touching other than stopping out pinholes should be permitted. If awards are made—and in this I recall Colonel Wortley's mention of the medal of the Photographic Society—no award should be made for a touched negative unless a print from it before touching should be submitted *to the judges* at the same time, but not necessarily for exhibition.

It is clear that when we talk of excellence in photography we mean something other than what a draughtsman may do in the way of supplementing photography. Works which do not enter for an award may omit mention of the distinction I have indicated.

I believe that such an exhibition would excite a very general interest, and do more to improve the knowledge of photographers on practical details than years of casual acquaintance of what other men do.

I am, to a certain extent, an outsider, and cannot do more than suggest; but I hope that some of the influential masters of the camera will take up the question.—I am, yours, &c.,

W. J. STILLMAN.

100, Clarendon-road, Notting-hill, January 29, 1872.

NATURAL COLOURS IN PHOTOGRAPHY.

To the EDITORS.

GENTLEMEN,—Last night, as “I lay a-thinking,” the subject of obtaining photographic colour suddenly occurred to me, and the question arose as to whether the differences in the colours of the light reflected from the surfaces of different-coloured flowers (as red, blue, and yellow) was due to differences in the constitution of their juices, or of the solid matter of which they are formed. If on examination it proved to be of the juices, a second question arose as to whether it would not be possible to take advantage of this in the preparation of plates, as suggested by me in an article which you inserted in *THE BRITISH JOURNAL OF PHOTOGRAPHY* for October 27, 1865 (No. 286, vol. xii.). Then, supposing that by this or any other means the three monochromic plates were obtainable, would coloured glasses placed in front of the lens help to stop off the colours not to be represented on the plate? Thus, with a plate sensitive to blue only, the interposition of a blue glass would prevent the transmission of the yellow and red rays, a red glass those of the blue and yellow rays, and a yellow glass that of the blue and red rays to the plates sensitive to the blue, red, and yellow rays respectively.

I have evidence that so early as 1842 the late Sir John Herschel obtained variously-coloured photographs on paper, as he gave me several, and I have still one a good blue, one a fair red, and two purple. The letter accompanying them describes the two latter as produced by the use of the juice of the red poppy; but many of them have faded away entirely. My impression is that others of them were from vegetable juices, but I am not sure that this was the case. There can be no doubt, however, that his published papers will give an account of the numerous experiments he made on this subject.

These suggestions may, or may not, be of value in forwarding the realisation of the great *desideratum* of photographic colour; but I cannot be far wrong in mentioning them as they occur to me, especially as my former communication, in 1865, was thought to be worth consideration by experimentalists.

That the great end will be attained before a very long period has elapsed, and the prediction of M. Niepce be verified, that “one day a photographic picture will be produced such as one sees in a looking-glass,” is the hope and wish of—Yours, &c.,

HENRY COLLEN.

Milford, Godalming, January 29, 1872.

THE PHOTOGRAPHIC DEPARTMENT OF THE
INTERNATIONAL EXHIBITION.

To the EDITORS.

GENTLEMEN,—The notice which the direction of the International Exhibition has sent to your Journal is sufficiently unsatisfactory to be discussed a little before the photographic profession commits itself to the mercies of that institution for another such display as we had last year.

It seems to some of us that the least the management could do, if the leading photographers are expected to contribute, would be to put some one on the photographic committee whom they are accustomed to regard as identified in a high degree with the interests of the art, or whose interest in it they feel assured of. We should have imagined that one of the presidents of the photographic societies, or at least one of those eminent amateurs who have really contributed to the advancement of photographic science, and shown a disinterested devotion to it in its present condition, would have had the selection, or, at least, a voice in the selection, of the pictures to be exhibited.

As it is, we have Dr. Diamond, who was, in years gone by, interested in photography, and who is understood to be in the present combination a passive member; Mr. Thompson, of whom most of us know nothing; and Col. Stuart Wortley, whom some of the profession do not accept as an authority, and in whose position, as having a commercial interest in photography in no way identified with that of the profession at large, they find excellent reasons why he should not be put forward as the judge and spokesman of it. If a professional photographer is to be assigned this position, Col. Wortley's place is not sufficiently high to justify his selection. If an outsider must be selected, he is disqualified, as being commercially interested on the one hand and a disputed authority on the other, or one at least to whom few good professionals will defer.—I am, yours, &c.,

HYPPO.

London, January, 29, 1872.

[Endorsing all that is said about the two jurors first mentioned, we ask our correspondent if he can, after due consideration, indicate any photographer, professional or amateur, in whom a greater degree of confidence would be placed than in Col. Wortley?—EDS.]

COMBLIKE MARKINGS ON THE NEGATIVE.

To the EDITORS.

GENTLEMEN,—Perhaps the following may explain the defect “J. H. M.” speaks of:—During the hot weather last summer, while photographing an engine in the open air, every plate showed comblike marks at one side of the plate. After two days’ trial, and filtering the bath, changing the collodion, &c., it struck me that the cause might be the partial drying of the film. A piece of wet blotting-paper at the back of each plate at once remedied the defect, and all came right. I now always place wet paper at the back under similar circumstances.—I am, yours, &c.,

GEO. SPENCER.

77, Cannon-street, E.C., January 26, 1872.

AMMONIA FUMING.

To the EDITORS.

GENTLEMEN,—In your useful little ALMANAC for the present year there is an article by Mr. A. L. Henderson, enthusiastically written, in favour of fuming.

Now, in accordance with his recommendation, I have tried the said fuming; but my first essay has certainly not impressed me very strongly as regards its favourable results—whether from my own defective manipulation or not I cannot say.

I found that the paper—especially some of Marion's thick Saxe—assumed a most disagreeable yellow colour after fuming, and it maintained that colour through all subsequent operations—as a matter of course, spoiling the prints. Another sample of (Rive) paper procured from another dealer was not quite so faulty in this respect; still, in neither case was the brilliancy of the prints enhanced—rather deteriorated, I thought—although Mr. Henderson maintains so stoutly the advantages of fuming in this respect. Again: instead of the toning being quicker and more regular in action, as Mr. Henderson states, I found it much the reverse.

I think that I must be wrong somewhere in my working, and I wish to ask your opinion or Mr. Henderson's on the best mode of proceeding—assuming it to be really worth while to adopt fuming. I used an oblong box about two feet in length by one foot in breadth, placing at the bottom a narrow-necked bottle containing about two ounces of liquor ammonia, fuming two quarter-sheets of paper for about ten minutes, with the result above stated.

Have I omitted any necessary addition? or have I proceeded wrongly?—I am, yours, &c.,

AMMONIA FUMING.

Leeds, January 29, 1872.

A CORRECTION.

To the EDITORS.

GENTLEMEN,—In my communication to the Journal last week, at page 37, under heading *Using the Substratum*, there is an omission which would prevent the successful coating of the plates.

The strength of the albumen there given is that of the stock solution, to use which *add two ounces of distilled water to every half-drachm required*, half-a-drachm thus diluted being amply sufficient to coat fifty plates.—I am, yours, &c.,

ALECK. A. INGLIS.

Edinburgh, January 30, 1872.

Miscellanea.

TRADE IN OBSCENE PHOTOGRAPHS.—A man named Benjamin Smith, *alias* Martin Stanley, living at Oil Mill Folds, or Alma-place, Westgate, Rotherham, was charged at Rotherham, on Thursday, the 25th ult., with publishing and selling photographs of a kind to debase and scandalise human nature. The West Riding constabulary received information from various sources that photographs of a most disgusting character were being circulated throughout the United Kingdom, and even in other countries, by a man living at Rotherham, who had inserted an advertisement in several London and provincial papers, in the name of Smith and Stanley, stating that he was prepared on receipt of postage stamps to forward portraits of French girls of a highly novel and exciting nature. A trap to catch the prisoner was accordingly laid by the Rotherham police, an inspector sending to him letters signed with an assumed name, and purporting to come from a place some distance from Rotherham. Stamps were enclosed, and in return a number of photographs of a most filthy and disgusting nature were obtained. At length some postage stamps, bearing a private mark, were sent in a letter to the prisoner, and the inspector of police watched the postman deliver it. He then entered the house, and found his own letter containing the marked stamps, together with a number of other letters, in possession of the prisoner. On the house being searched another beastly photograph was found, whereupon the prisoner told the officer that he would find no more there, as he kept them at a house he mentioned at Sheffield. This house was accordingly searched by two officers of the Sheffield police force, and there they found a number of most filthy photographs, together with printed lists describing what the prisoner had for sale. The prisoner was apprehended on the 14th of January, and since that time nearly a hundred letters from all parts of England, Scotland, and Ireland have arrived at the Rotherham post office directed to him. The police obtained the permission of Her Majesty's Secretary of State and of the Postmaster-General to receive and open these letters, and they found, on

doing so, that great numbers of postage stamps were enclosed, and some filthy photographs and books. The prisoner was committed for trial, and on application being made the magistrates intimated that they should require the prisoner to be bound in the sum of £200, and two sureties of £100 each to be found.

CAMPBOR IN THE PRINTING BATH.—Mr. John R. Clemons gives in *Mosaics* some of his experience with camphor in the printing press. He says:—After the positive printing bath has been used some time it becomes more or less charged with albumen. If, when the solution is poured from the dish into the filter, frothy bubbles appear on the surface, it is in the condition named, and is unfit for use in sensitising albumen paper. The reason is obvious. When an albumen sheet is floated upon a silver solution thus charged with albumen, a secondary film of albumen, or albumenate of silver, is imparted to it, which deprives it of its lustre, and it is impossible to secure a good tone upon it. The difference will be readily seen by floating one-half of a sheet of paper on the used solution, and the other half on a fresh solution, and comparing the results. The effect of silvering on such a solution is similar to that of using doubly-albumenised paper. In both cases the silver will penetrate both films, even if silvered on the back, and very good prints apparently may be made on paper so treated; but when you proceed to tone them, you will find that the double film of albumen resists the action of the gold, and renders it impossible to get rich tones. I have already recommended the addition of alcohol to a bath thus charged with albumen, and then burning it out, in order to cleanse the bath, but since have sought for a less expensive method, and have found camphor to be just what is needed. Besides being less expensive, it is also more expeditious; for in five minutes an eighty-ounce bath can be cleansed of all impurities by its use, as directed below. Make a saturated solution of camphor, viz.:—

Camphor 1 ounce.

Alcohol 6 ounces.

To cleanse a bath which is considerably fouled by the albumen, add two and a-half ounces of this camphor solution. A greasy appearance will be presented on the surface of the bath. Shake well, when the greasiness will disappear. Then filter, never using the same filter twice. If, after filtration, the solution turns dark, add a couple of drops of permanganate of potash,

and it will immediately clear. This turning dark is owing to long usage of the silver solution. It is a fact that the bath is daily impregnated more or less with the albumen, therefore a slight addition of camphor daily is recommended. This will avoid the addition of the permanganate, which rather decreases the sensitiveness. As a quick and ready corrective agent, camphor will be found of great advantage. It will impart a camphory smell to the solution, but in no way deteriorates the quality of the prints.

TO COPY OIL PORTRAITS.—Mr. B. Frank Saylor, in *Photographic Mosaics*, says on this subject:—As I have never seen anything from which I have derived any material aid in the copying of oil portraits, and especially old ones, I propose simply to give our *modus operandi*, feeling confident that those who do not proceed in like manner may successfully copy oil paintings, and especially old portraits. We first, with a clean sponge and water, partially wash the old portrait, and then pour perhaps an ounce of glycerine on it, and with the same sponge brush the glycerine crosswise over the entire picture until it presents a uniform surface; or, should a greasy appearance be presented, something like a negative plate when partially coated, rub gently with the palm of the hand crosswise, and it is ready for copying. We copy under the skylight, about where we should place the sitter for the same lighted picture, directing the camera to about the centre of the picture, and as nearly level as may be. On our camera we use a quarter-size tube inverted—that is, turn it end for end—with a diaphragm of about six-eighths of an inch aperture, placing the picture square to the front of the camera. After making the necessary adjustments as to size, &c., we coat the plate with the same collodion which we use for our regular portrait negatives, and when it is properly coated expose fully—that is, neither over- nor under-expose. The success, however, depends on the manipulation *during the exposure*. We take a piece of backboard, say six inches wide and about three feet long, and with it shade all the top of the portrait above the head, standing by the side of the portrait and holding the backboard in the hand, moving it up and down as though it were hinged on the top of the portrait, but not allowing its shadow to come down over the forehead or face, except those parts should be improved thereby. We produce the same effect on the sides of the portrait by occasionally moving our body at the same time over portions of the background, taking care not to remain perfectly still or too long in the same place, nor yet to allow any shadow from our person or anything else to cover any portion of the figure.

Thus, with a little care and attention, a very satisfactory background is obtained, even admitting of desirable variation, thereby avoiding the necessity of a double-printed background and the like, and producing from a single negative and printing duplicates to equal, if not to surpass, the original, and every print alike. Lastly: we develop with the usual iron developer, using six ounces of iron to sixty-four of water; this, however, we again dilute with water, adding alcohol, more or less acetic acid, according to the detail and density we wish to obtain.

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London: HENRY GREENWOOD, publisher, 2, York-street, Covent-garden, W.C.

EXCHANGE COLUMN.

No charge is made for inserting these announcements; but in no case do we insert any article merely *offered for sale*, that being done at the small cost of one shilling in our advertising pages. This column is devoted to exchanges only. It is imperative that the name of the person proposing the exchange be given (although not necessarily for publication, if a *nom de plume* be thought desirable), otherwise the notice will not appear.

I will exchange a Marion's embossing press for cameo *cartes*, cost £4 10s., for anything useful in photography. Offers invited.—Address, T. KAY, Lark Hill House, Bolton.

A very superior 36-inch photographer's bicycle will be exchanged for a fine collection of lantern slides, revolving stereoscope, or other photographic materials. Offers solicited. Photograph and particulars on application.—Address, H. ROGERSON, 10, Springcliff, Bradford.

I wish to exchange an exterior background, by Bull, cabinet size, for another, or anything useful for the studio; it was new last June, and cost 37s. 6d. Offers also wanted for a single half-plate view lens, by Home and Thornthwaite. Photographs exchanged.—Address, Photographic Institute, Portmadoc.

ANSWERS TO CORRESPONDENTS.

✉ *Correspondents should never write on both sides of the paper.*

JAMES TULLEY.—Received. In our next.

GEO. DANIEL.—Professor Wheatstone was the inventor of the reflecting stereoscope. An account of his invention was published in 1838 in the *Philosophical Transactions*.

H. W.—The *carte* enclosed is a very pretty one, both in respect of the subject and the treatment. Had the oval been omitted and the dome alone used the effect would have been better.

PHOTO.—Let the south side of the studio be glazed—not with clear but with ground glass. Should this not be procurable in your town, use plain glass, and dab it over with putty.

CRAWDON.—India-rubber finger-stalls will prove an effectual means of keeping fingers clean. We do not use them ourselves; but we know several gentlemen who do. They cost about threepence each.

EAST ANGLICAN.—There is no remedy for the spots in the paper except more care in the selection. The spots are owing to minute particles of metal introduced into the paper when in a state of pulp.

T. M. (Newcastle-on-Tyne).—So far as we know, the only house in the photographic trade from which you can obtain what is wanted is that of Mr. F. W. Hart, 8, Kingsland-green. The other matter shall receive attention.

W. W.—1. A vessel made of gutta-percha will both destroy and eventually be destroyed by a solution of nitrate of silver.—2. Slate does not appear to affect a nitrate bath.—3. Bees' wax is not decomposed by this solution.

PIETRO CONSTANTINO REMONDINI (Genoa).—Enclosure received with thanks.
The ALMANAC was posted on the 22nd January.

G. W. H.—It is quite impossible from such slender data as that supplied by you to obtain the information desired, but from the nature of the subject we are strongly of opinion that it would be dangerous for you to follow out your proposed intention.

INQUISITIVE READER.—You are mistaken in your supposition concerning the author of the article. He is not an amateur, but a professional of high standing, who, for certain reasons, prefers the adoption of a *nom de plume* to giving his own name.

INQUIRER.—We could not in this column give you a proper idea of polarised light. The “three words” into which you would like the information compressed would have to be extended to many more before we gave you or our readers a clear idea of the nature and properties of polarised light.

PHOTOLITHO.—We are aware that Mr. Woodbury has so altered his process as to get rid of the necessity for using a preliminary coating of collodion. He now prints on the bare gelatine, and after exposure mounts the film, sensitive side downwards, on a plate of glass coated with gelatine and chrome alum.

PORTRAITS BY THE MAGNESIUM LIGHT.—Mr. W. G. Lewis has sent us some *carte* portraits taken at night by the magnesium light, which show that in his hands this neglected branch of photography is very successfully managed. The faces are full of delicate half-tone. The time of exposure was three times that required in ordinary daylight.

GEORGE THOMPSON.—1. Although we do not esteem the maker of your lens as one of the best, he has, for all that, a fair reputation in his own country; and we know that he has made some good lenses. We have not, however, heard of him for several years past.—2. Dr. Monckhoven is a dealer in apparatus—not a manufacturer of lenses.

F. C. S.—Ideas often run in parallel grooves. For several weeks past we have been using our note-book very freely in the collection of materials for such a series of articles as that suggested by you. Some of the information required to render the series of the greatest possible value is only procurable with much difficulty and at no little expense; still, such

progress in the compilation has been made as will warrant the indulgence of a hope that we may be able to introduce the subject at no distant date.

ERRATUM.—In Mr. Coote's article in our last number, page 38, second column, eleventh line from top, for "you are apt to hurry and *free* the development," *read* "FORCE the development." Mr. Coote also informs us that in the middle paragraph of the first column in the same page, a sentence there in which he gave his reason for discontinuing the use of the salt bath for washing the plates will read better and more correctly as follows:—"Firstly: I found the extra salt bath and long washing required after it to considerably lengthen an already sufficiently tedious method of preparation."

SUBSCRIBER.—Excuse plain speaking, but your pictures are by no means good—in fact, they are *very* bad. We are at a loss as to what to attribute their special qualities—whether to want of care or want of knowledge and experience. Carelessness, to all appearance, has had much to do with the matter; for the plate has been badly coated and is torn, and the developer has not flowed smoothly over the surface. After you have acquired some more experience and dexterity in manipulation write again, enclosing a specimen. You need not entertain any fear of your name and address being divulged.

W. K.—No more easy and expeditious method of collecting oxygen in a bag can be adopted than that which has so often been described in our Almanacs and Journals, viz., connecting the bag with the washing bottle by means of an india-rubber tube, the bottle being in turn connected with the retort in which the oxygen is generated. The method of making oxygen from chlorate of potash and black oxide of manganese is the simplest, cheapest, and best to adopt when only a small quantity is required. By a small quantity we here mean such a quantity as will be required by a professional enlarger of photographs in full practice.

SAM WELLER says—"1. Can you inform me what the method is for enlarging designs for calico or zinc plates to be engraved for the purpose of tracing from by pentagraph?—2. Also as to the manner of building a suitable glass house, and kind of light—north or south?—3. Also, if a cheap condenser for the solar camera may be made by cementing a clock glass to a plate of sheet glass and filling it with water?"——We reply:—

1. A design can be placed upon a zinc plate by the combination of enlarging and photolithography, or by transferring a carbon enlargement.—2. A north light is better than a south; for the rest, it will be necessary to consult an architect.—3. This query is answered with some degree of minuteness in the first paragraph of page 30 of our ALMANAC for the present year.

OLD HARRY asks—“What is a pistolgram?” We are to some extent ashamed of the definition we are about to give, but we can offer no other. It is a small picture taken by a *pistolgraph*, and this, in turn, is a small camera introduced by Mr. Thomas Skaife, by which photographs might be taken with a very rapid exposure. We strongly suspect that “Old Harry” knows more of the subject than he wishes us to believe, otherwise why put the question concerning the greater angularity of aperture that is possible to be obtained with a short than with a long-focus lens? and to which we reply that, without at present speculating on the cause, the effect is just as he states it to be. The same principle applies to the object-glasses of microscopes, and it is an important element in the recognition of object-glasses of the highest quality.

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- Transcriber's Notes:

- Missing or obscured punctuation was silently corrected.
- Typographical errors were silently corrected.
- Inconsistent spelling and hyphenation were made consistent only when a predominant form was found in this book.

*** END OF THE PROJECT GUTENBERG EBOOK THE BRITISH
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